



# Business Understanding Report

Team Innovators - DEPI CAPSTONE PROJECT - UK Train Rides

## 1. Business Problem Statement

### PS 01:

The company wants to analyze **service delays** to identify the main reasons customers request ticket refunds in order to reduce financial losses caused by delays.

### PS 02:

The company wants to analyse **ticket pricing** and **purchase patterns** to evaluate the impact of the 50% advance-purchase discount on revenue, since a large percentage of customers buy tickets in advance, and the high discount may reduce overall profitability.

### PS 03:

The company wants to analyse ticket purchases at stations and purchase times to identify **peak station** demand periods in order to reduce the crowding, and evaluate the ticket price.

## 2. Business Processes

A customer selects a route (depart → arrive station) and ticket type (Advance - First Class - Anytime) and completes the purchase either online or at the station.

The system records the transaction and schedules the journey for a specific date. On the (Date of Journey), the actual arrival time is compared against the scheduled time.

If a disruption occurs (delay - canceled), the customer may then initiate a "Refund Request" through the company's customer service process.

## 3. Stakeholders

| Stack Holder                | Interest                                                                   | Type     |
|-----------------------------|----------------------------------------------------------------------------|----------|
| <b>Sales Management</b>     | Accurate pricing strategies to maximize revenue.                           | Internal |
| <b>Operation Management</b> | Reducing operational pressure and optimizing workflow efficiency.          | Internal |
| Marketing Team              | Fine-tuning discount structures and promotional campaigns.                 | Internal |
| Customer Service            | Reducing refund requests and improving satisfaction for unhappy customers. | Internal |
| Government                  | Compliance with regulations, taxation, and legal frameworks.               | External |
| <b>Passengers</b>           | Reliable service, fair pricing, and a smooth user experience.              | External |

## 4. Analytical Questions & KPIs

### Sales Management

#### KPI

- Gross Sales
- Net. Revenue (Revenue after refunds)
- Avg. Ticket Price
- Discount Rate %
- Price Realization % ( $\text{Price} / \text{Price before discount}$ )
- Total Loss from Refund (Refund Amount)
- Total Loss from Discount (Discount Leak)

#### Questions

- What is the highest-revenue Purchase Type, and does it differ in average ticket value?
- Revenue mix by Ticket Type (Advance/Off-Peak/Anytime)
- **What % of total revenue comes from the top 5 lines?** (Concentration risk)
- **Which Railcard × Ticket Type combination drives the highest discount leakage relative to its revenue share?**
- Revenue per ticket Classes
- Refund Impact on Revenue (Trend)
- Revenue over time
- Top 5 high-revenue lines

### Operations

#### KPI (per day)

- Refund Request Rate %
- Delay Rate %

- On Time Rate %
- Cancellation Rate %
- Peak Hour
- Total Journeys
- Avg. Delay Mins

## Questions

- What are the **reasons for the delay** that make customers **request a refund**?
- At what times do the **reasons for delay** affect the **journey delay**?
- Top 5 high-demand lines
- What is the busiest (time of day - day of the week) (peak time)
- For each reason for delay, Avg. Delay Time.
- What is the expected journey time for each line?
- What is the estimated time between ticket purchase and journey time per Ticket Type?
- What is the most common purchase type?
- What are Departure Time Peak Hours Range?

## Passengers & Marketing

### KPI

- Total Passengers
- Total Passengers have a Railcard
- Special Need Percentage
- Avg. Ticket Price

## Questions

- What is the Distribution of person who holds a Railcard (Adult - Senior - Disabled) (Pie Chart)
- Purchase Method (Contactless - Debit)
- Channel (Sales - Revenue - Volume) rate between online and station.

- Which Ticket Type/Railcard combination is most associated with each Purchase Type?
- Is there a relationship between Payment Method and Refund Request rate?

## 5. Data Sources

| Data Source          | Description                                 | Columns                                                              |
|----------------------|---------------------------------------------|----------------------------------------------------------------------|
| Railway Transactions | Individual train ticket purchase records    | Date of Purchase Time of Purchase Purchase Type Price Refund Request |
| Stations             | Departure and arrival station information   | Departure Station Arrival Destination                                |
| Journey              | Scheduled and actual journey timing details | Date of Journey Departure Time Arrival Time Actual Arrival Time      |
| Ticket               | Ticket class, type, and pricing information | Ticket Class Ticket Type                                             |
| Passenger            | Railcard and payment method details         | Railcard Payment Method                                              |
| Delay                | Journey status and reasons for delay        | Reason for Delay Journey Status                                      |

## 6. Data Dictionary

| Field                   | Description                                                            | Data Type         | Example                 |
|-------------------------|------------------------------------------------------------------------|-------------------|-------------------------|
| <b>Transaction ID</b>   | Unique identifier for an individual train ticket purchase              | CHAR(23)          | da8a6ba8-b3dc-4677-b176 |
| <b>Date of Purchase</b> | Date the ticket was purchased                                          | Date (YYYY-MM-DD) | 2023-12-08              |
| <b>Time of Purchase</b> | Time the ticket was purchased                                          | Time (HH:MM:SS)   | 12:41:11                |
| <b>Purchase Type</b>    | Whether the ticket was purchased online or directly at a train station | VARCHAR(255)      | Online, Station         |

| Field                      | Description                                                                                                                                                                                                                                                                                                               | Data Type         | Example                                |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------|
| <b>Payment Method</b>      | Payment method used to purchase the ticket (Contactless, Credit Card, or Debit Card)                                                                                                                                                                                                                                      | VARCHAR(255)      | Contactless , Credit Card , Debit Card |
| <b>Railcard</b>            | Whether the passenger is a National Railcard holder (Adult, Senior, or Disabled) or not (None). Railcard holders get 1/3 off their ticket purchases.                                                                                                                                                                      | VARCHAR(255)      | Adult , Senior , Disabled , None       |
| <b>Ticket Class</b>        | Seat class for the ticket (Standard or First)                                                                                                                                                                                                                                                                             | VARCHAR(255)      | Standard , First Class                 |
| <b>Ticket Type</b>         | When you bought or can use the ticket. Advance tickets are 1/2 off and must be purchased at least a day prior to departure. Off-Peak tickets are 1/4 off and must be used outside of peak hours (weekdays between 6-8am and 4-6pm). Anytime tickets are full price and can be bought and used at any time during the day. | VARCHAR(255)      | Advance , Anytime , Off-Peak           |
| <b>Price</b>               | Final cost of the ticket                                                                                                                                                                                                                                                                                                  | Integer           | 43                                     |
| <b>Departure Station</b>   | Station to board the train                                                                                                                                                                                                                                                                                                | VARCHAR(255)      | London Paddington                      |
| <b>Arrival Destination</b> | Station to exit the train                                                                                                                                                                                                                                                                                                 | VARCHAR(255)      | Liverpool Lime Street                  |
| <b>Date of Journey</b>     | Date the train departed                                                                                                                                                                                                                                                                                                   | Date (YYYY-MM-DD) | 2024-01-01                             |
| <b>Departure Time</b>      | Time the train departed                                                                                                                                                                                                                                                                                                   | Time (HH:MM:SS)   | 11:00:00                               |
| <b>Arrival Time</b>        | Time the train was scheduled to arrive at                                                                                                                                                                                                                                                                                 | Time (HH:MM:SS)   | 13:30:00                               |

| Field                      | Description                                                                   | Data Type        | Example                                        |
|----------------------------|-------------------------------------------------------------------------------|------------------|------------------------------------------------|
|                            | its destination (can be on the day after departure)                           |                  |                                                |
| <b>Actual Arrival Time</b> | Time the train arrived at its destination (can be on the day after departure) | Time (HH:MM:SS)  | 13:30:00                                       |
| <b>Journey Status</b>      | Whether the train was on time, delayed, or cancelled                          | VARCHAR(255)     | On Time , Delayed , Cancelled                  |
| <b>Reason for Delay</b>    | Reason for the delay or cancellation                                          | VARCHAR(255)     | Signal Failure , Weather Conditions , Staffing |
| <b>Refund Request</b>      | Whether the passenger requested a refund after a delay or cancellation        | Boolean (Yes/No) | Yes , No                                       |

## 7. Assumptions and Constraints

### Data Cleaning Transformations

- Missing **Railcard** values in raw data (no railcard held) were recoded as "Null" example refers to the raw/pre-cleaning state).
- Missing **Reason for Delay** values occur only for "On Time" journeys (logically expected) and were filled as "No Delay".
- Inconsistent raw labels in **Reason for Delay** were standardized to one canonical label each (e.g., "Signal failure"/"Signal Failure" → "Signal Failure"; "Staffing" → "Staff Shortage"; "Weather" → "Weather Conditions").
- **Actual Arrival Time** is missing by definition for all Cancelled journeys (the train never ran) — not treated as a data quality issue, and excluded from delay-time calculations for that status.

### Known Data Quality Issues

- `Arrival Time` / `Actual Arrival Time` can fall on the day after departure (overnight services). Journey duration and delay calculations must account for day-rollover; otherwise ~3% of journeys (overnight routes) produce invalid negative durations.
- "Edinburgh" and "Edinburgh Waverley" appear as two distinct values in `Arrival Destination` for what is the same physical station — Route-level aggregations (Top N lines, revenue per line) will undercount this destination unless merged.

## Business Logic Assumptions

- `Refund Request = Yes` is treated as a successful/full refund of `Price` — the source data has no separate "Refund Approved/Rejected" status, so "Total Refunded Amount" assumes every request was honored in full.
- `Price` reflects the amount actually charged (post-discount). `Price before discount` and `Total Discount %` are reverse-derived by assuming Ticket Type discount and Railcard discount combine **multiplicatively** (e.g., Advance + Railcard =  $1 - (0.5 \times 0.667) \approx 66.7\%$  off), since the business has not explicitly confirmed whether discounts stack multiplicatively or additively.